

# ML Lead Interview Questions

## Mapped to Job Description Requirements

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### 1. ML Model Development & RAG Pipelines

**Q1:** You're asked to build an enterprise RAG system for a company with 500K+ internal documents across SharePoint, Confluence, and PDFs. The system must handle 5000 daily queries from business users. How do you design this end-to-end?

**What to listen for:**

- Chunking strategy for different document types
- Embedding model selection (open-source vs proprietary)
- Vector DB choice with enterprise-grade security
- Hybrid retrieval (semantic + keyword)
- Access control and permission-aware retrieval
- Scalability and cost planning

**Red Flag:** Only talks about the model, ignores enterprise concerns like security, permissions, and document freshness.

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**Q2:** Your deep learning model works great in research but fails to scale for production business needs. How do you bridge the gap between experimental ML and scalable business solutions?

**What to listen for:**

- Model compression/distillation for scalability
- Decoupling model logic from business logic
- Serving infrastructure (TorchServe, TF Serving, FastAPI)
- Batch vs real-time inference decisions

- Business metric alignment (not just model metrics)
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## 2. Model Training, Evaluation & Python/Microservices

**Q3:** Walk me through how you would design a robust, reproducible model training pipeline that integrates with a microservices architecture. A new dataset arrives every week.

**What to listen for:**

- Orchestration tools (Airflow, Kubeflow, Prefect)
- Experiment tracking (MLflow, W&B)
- Data versioning (DVC)
- Containerized training (Docker)
- API design for model serving (FastAPI/Flask)
- Event-driven triggers for retraining
- Decoupled microservices for preprocessing, training, evaluation, and serving

**Red Flag:** Describes a monolithic notebook-based pipeline with no automation.

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**Q4:** How do you evaluate a model beyond standard metrics like accuracy or F1? Give an example where standard metrics were misleading and how you caught it.

**What to listen for:**

- Business metric alignment (revenue, churn, CSAT)
  - Slice-based evaluation (performance per segment)
  - Behavioral testing / adversarial test cases
  - Real-world vs offline metric gap
  - Calibration checks
  - Fairness and bias audits
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## 3. Feature Engineering & Cloud Deployment

**Q5:** You are inheriting a feature engineering pipeline that takes 6 hours to run daily. It has no documentation and breaks frequently. How do you modernize it while keeping production stable?

**What to listen for:**

- Audit and documentation first approach
  - Incremental migration (don't rewrite everything at once)
  - Moving to a feature store (Feast, Tecton, SageMaker Feature Store)
  - Parallel processing (Spark, Dask, Ray)
  - Unit testing for feature transformations
  - Monitoring for feature drift
  - Zero-downtime migration strategy
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**Q6:** You need to deploy an ML model across AWS for one business unit and Azure for another, due to enterprise compliance requirements. How do you design for this multi-cloud reality?

**What to listen for:**

- Cloud-agnostic abstraction layers
  - Containerization with Docker/Kubernetes for portability
  - Infrastructure as Code (Terraform)
  - Using managed services: SageMaker (AWS), Azure ML, Vertex AI (GCP)
  - CI/CD that targets multiple cloud environments
  - Cost monitoring per cloud
  - Data residency and compliance awareness
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## 4. MLOps & CI/CD

**Q7:** Describe the most mature MLOps pipeline you have built or led. What did it look like at Level 0, and how did you get it to Level 2 or 3 maturity?

**What to listen for:**

| Level   | Description                                    |
|---------|------------------------------------------------|
| Level 0 | Manual, notebook-based, no automation          |
| Level 1 | Automated training, manual deployment          |
| Level 2 | Automated training + deployment pipeline       |
| Level 3 | Full automation with self-triggered retraining |

- Tools used: Jenkins, GitHub Actions, GitLab CI, Kubeflow
- Model registry and versioning
- Automated testing for data, model, and infrastructure
- Rollback mechanisms
- Governance and audit trails

**Red Flag:** Has only experienced Level 0 or cannot articulate the progression.

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## 5. Model Deployment & Production Reliability

**Q8:** One of your production models serving 50,000 requests/day suddenly degrades in performance on Monday morning. You have no alerts set up. Walk me through everything — detection, diagnosis, fix, and prevention.

**What to listen for:**

**Detection:**

- How they'd notice without alerts (user complaints, dashboards)
- Immediate triage steps

**Diagnosis:**

- Data pipeline failure vs model drift vs infrastructure issue
- Checking input feature distributions
- Comparing recent vs historical predictions

**Fix:**

- Rollback to last stable model version

- Hotfix vs proper fix decision

**Prevention:**

- Monitoring setup: data quality, prediction distribution, business KPIs
  - Alerting thresholds
  - Shadow mode deployment
  - Automated retraining triggers
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## **6. Collaboration, Leadership & Mentoring**

**Q9:** You have a junior ML engineer who consistently submits models that perform well in notebooks but fail in production. How do you coach them without micromanaging, and what systems do you put in place?

**What to listen for:**

- Structured code review process with ML-specific checklist
  - Pair programming on production deployment tasks
  - Creating internal standards and templates
  - Teaching mindset (root cause, not just fixing symptoms)
  - Psychological safety — not shaming but learning
  - Measurable improvement goals
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**Q10:** You are leading a project that requires input from Data Science, Product, and Engineering teams. The teams have conflicting priorities and the project is at risk of delay. How do you get alignment?

**What to listen for:**

- Stakeholder mapping upfront
- Establishing shared success metrics (not team-specific)
- Escalation path when blocked
- Ruthless prioritization of MVP vs nice-to-haves

- Structured communication (weekly syncs, async updates)
- Owning outcomes, not just tasks

**Red Flag:** Blames other teams or says “I just focus on the ML part.”

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## 7. Innovation & Research Awareness

**Q11:** In the last 6 months, what is one new ML research paper, framework, or tool you explored and applied (or recommended) at work? What was the outcome?

**What to listen for:**

- Specific and genuine answer (not vague)
- Ability to translate research into business value
- Papers/tools relevant to current GenAI landscape (e.g., RAG improvements, LLM efficiency, multi-modal models)
- Critical thinking — also able to say “we evaluated it but it wasn’t ready”

**Red Flag:** Cannot name anything specific or only mentions very old tools/concepts.

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## Evaluation Scorecard

| Area                        | Question | Weight |
|-----------------------------|----------|--------|
| ML Model Development & RAG  | Q1, Q2   | High   |
| Training & Evaluation       | Q3, Q4   | High   |
| Feature Engineering & Cloud | Q5, Q6   | Medium |
| MLOps & CI/CD               | Q7       | High   |
| Production & Deployment     | Q8       | High   |
| Leadership & Collaboration  | Q9, Q10  | High   |
| Innovation & Research       | Q11      | Medium |

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## Key Things to Look for in an ML Lead

- ✓ Thinks about **business impact**, not just model performance
- ✓ Has **production scars** — knows what breaks and why
- ✓ Can **communicate** complex ML to non-technical stakeholders
- ✓ **Grows the team**, not just does the work themselves
- ✓ Makes **pragmatic trade-offs** between perfection and delivery
- ✓ Has hands-on **cloud and MLOps** experience, not just theory
- ✗ Only talks about algorithms and ignores deployment
- ✗ Cannot give specific real-world examples
- ✗ Blames other teams for failures
- ✗ Has no opinion on trade-offs — agrees with everything